Catalyst project #1200085 closeout report

Name: Conditional Transfer of Digital Assets via Digital Identity and Verified Credentials

Ideascale: https://cardano.ideascale.com/c/cardano/idea/120430

Milestones tracking: https://milestones.projectcatalyst.io/projects/1200085

Project Mgr: Eric Duneau

Started on: 12 August 2024

Completed on: 15 January 2024

Amount: 32,000 ADA

Challenges:

- The use of Identus is always a little bit of a challenge. I have had to install a full Identus agent node on AWS, and in the middle of the PoC, it stopped validating credentials for unexplainable reasons. After a few days of many attempts and still no solution, I decided to reinstall from fresh on another machine, and re-create the whole environment. It worked. I do not feel it is right for devs to have to do so much config and system maintenance, and Identus should come out of the box ready for use, in its latest release, and as a free multi-agents sandbox.

- There were many challenges and learnings in terms of designing the right solution for the encryption and avoiding databases as much as possible. It is always the case when trying to deliver new solutions to market.

KPIs:

- The project was “mostly” kept on time, although I did not work on it for nearly 6 weeks, as I had over commitments in parallel. As always, getting back and forth between different projects can cost in time for re-adaptation and focus to the deliveries.

- I am overall very pleased with the architecture, and final solution. At the start, I did not expect that I would come up with such a final elegant solution, which is a very good base to reuse for building a final product. In this sense, the quality of the final delivery was achieved.

Achievements:

- The PoC works on top of an Identus cloud agent, but can also work independently of Identus, which is a great bonus.

- The PoC was written in Python, and can be found on the GitHub repo https://github.com/incubiq/one_crypto_pass - It is somehow unusual to have a Python library on top of Identus, but I made this choice when testing/dealing with various crypto libraries, and kept it all along.

- The high-level final design of the solution is documented here: https://github.com/incubiq/one_crypto_pass/blob/main/research/final_design.png

- We have an elegant solution for encoding / decoding secrets, and for sharing the conditional right to decode with third parties.

Learnings:

- I am becoming a top expert in Identus.

- I have a good grasp on encryption / cyber security, even if this is a "for-ever" evolving field.

Comments:

- it is my belief that based on this extensive PoC, we could get a final product to production for ensuring transfer of conditional assets within around 8 weeks.

- As a reminder, I did all this project alone, without any additional help (well... apart from my team of AIs who were incredibly good!)

Link to Repo: https://github.com/incubiq/one_crypto_pass

Link to all videos:

- M1 - Research + POC default solution: [16min video] https://www.youtube.com/watch?v=Qajs70mdwa0
- M1: Proof that Identus is fit for purpose: [30min video] https://www.youtube.com/watch?v=4DyPuZr_3PA

- M1: Conclusion / final chosen design: [11min video] https://www.youtube.com/watch?v=3rv1MJzEXGY

- M2: A PoC with live demo: [17min video] https://youtu.be/M4DpnHqJjOk

- Close Out: [5min video] https://youtu.be/WvPxJf8VHbk